

Mansour Doumbia

mansourdoumbia@college.harvard.edu | New York, NY | 347-364-4481 | mansourd.com

EDUCATION

Harvard University | Cambridge, MA

Expected 2028

Concentration: Mechanical Engineering, S.B.; Neuroscience Secondary

GPA: 3.7/4.0

Relevant Courses: Computer-Aided Design, Mechanical Systems, Thermodynamics, Control Systems, Space Science & Engineering

ENGINEERING EXPERIENCE

MIT Aero Astro | *Undergraduate Researcher* | Cambridge, MA

August 2026 – Present

- Developing a state-estimation system for encoderless field robots without built-in positional measurements using ultra-wideband (UWB) antennas.
- Integrating radio frequency (RF) sensing, modeling, and hardware testing to estimate joint states and accurate 3D positions.

Carillon Technologies | *Mechanical Engineering Intern* | New Haven, CT

May 2026 – August 2026

- Designed antenna test jigs and mounts using SolidWorks. Produced technical drawings, used FEA to verify displacement under compressive loading, and evaluated PCB manufacturability.
- Decreased sidelobe noise levels 8–10 dB and improved antenna gain 10–15 dB by building Python closed-loop VNA/gimbal software for $\pm 60^\circ$ azimuth/ $\pm 40^\circ$ elevation scans. Integrated genetic and covariance-matrix algorithms for optimal beam gain.
- Modeled liquid-crystal dielectric tensors across PCB structures in COMSOL, validating electromagnetic behavior and driving DFM/DFA changes for voltage-controlled beam pointing.
- Reduced 38,000-point Ka-band antenna scan from ~3 hours to ~1 hour by automating VNA data acquisition and gimbal motion while verifying gain and pointing against 2D radiation patterns.

NASA L'SPACE | *Mechanical Team, Administrative Lead* | Remote

August 2025 – May 2026

- Secured \$10k in funding and co-led an 8-person team to develop a 40-page mission concept selected for final review by NASA.
- Aided in design and prototyping of a multi-robot system, creating CADs with GD&T while integrating actuators and sensors.

Brothers Auto-Repair Shop | *Bodyworker & Customer Service* | New York, NY

September 2021 – September 2024

- Performed auto-body repairs, including MIG welding, panel alignment, and metal shaping to restore damaged vehicles.

ADDITIONAL EXPERIENCE

Harvard University Biorobotics Lab | *Undergraduate Researcher* | Cambridge, MA

August 2025 – January 2026

- Improved terrain classification accuracy from 85% to 92% for robotic navigation by developing a PointNet-based ML model.

Minerva.AI | *Co-founder and Head Designer* | New York, NY

May 2025 – September 2025

- Built an identity-verification and information-retrieval prototype with <300ms latency using Kotlin and C++ for XREAL AR glasses, integrating OpenAI Whisper and a TensorFlow/Keras Siamese network.
- Raised \$10k in pre-seed funding and engaged strategic stakeholders (including IQT) in early partnership discussions.

Harvard University Satellite Team | *Mechanical Subteam* | Boston, MA

January 2025 – May 2025

- Wrote NASA GMAT scripts to simulate satellite orbital and atmospheric trajectories, verifying mission feasibility and lifespan.

RELEVANT PROJECTS

Pulsejet Propane Engine

- Designed, manufactured, and bench-tested a propane-fueled pulsejet engine with a closed-loop propellant delivery system.
- Iterated combustion structure in 6 hot-fire tests, performing failure analysis to achieve exhaust temps of under ~600 °C.
- Used ANSYS and COMSOL to model acoustic resonance and pressure antinodes to drive chamber geometry changes.

Autonomous Control for Model Rocketry (Thrust Vectoring and Airbrakes)

- Designed and flight-tested embedded systems for reusable model rockets using PID and pressure-based altitude sensing.
- Integrated thrust vector control and deployable airbrakes to enable real-time autonomous trajectory correction and altitude control. Utilized a 20 Hz Pnut altimeter for telemetry, resulting in a ~10% improvement in mean altitude error across 20+ flights.

SKILLS

Mechanical: CAD (Fusion 360, SolidWorks, Onshape), GD&T, DFM/DFA PCB/PCBA, FEA, MIG Welding

Programming: Embedded Systems, C/C++, Kotlin, Python, Rust, MATLAB, Git

Analysis & Controls: COMSOL, ANSYS, LTspice, MATLAB, PID, State Estimation